Project Report - Day 10

Task-10: Benchmarking and Tuning the NexusVortex Knowledge Base

Objective

The objective of this task is to evaluate and improve the groundedness of a RAG-based technical Q&A system.
The system is tested using 5 fixed drill questions, scored on a 1-5 scale, and compared in a Before vs After experiment by tuning chunk size.

Libraries Used

1. **pathlib**
    - Used to load data.txt reliably from the script directory.
2. **os**
    - Used to read GEMINI_API_KEY from environment variables securely.
3. **sentence_transformers**
    - Used to load all-MiniLM-L6-v2 and create embeddings for document chunks and queries.
4. **sklearn.metrics.pairwise**
    - Used for cosine_similarity to retrieve the most relevant chunks.

- **google.genai**
- Used to generate grounded answers from retrieved context with strict prompt rules.

Implementation Summary

1. Loaded manual text from data.txt.
2. Created chunking pipeline using word-based chunking (avoids broken text fragments).
3. Built retrieval using query/document embeddings + cosine similarity.
4. Combined top-2 retrieved chunks as context for answer generation.
5. Applied strict grounded prompt: answer only from context, no hallucination.
6. Ran two configurations:
    - Before (Baseline): chunk size = 200
    - After (Tuned): chunk size = 600
7. Added score validation so only 1-5 is accepted.
8. Printed benchmark table with average groundedness scores.

Baseline Evaluation (Validated)

1. XV-505 question: 3/5
2. Lag question: 4/5
3. Theta-Sync / i7 question: 3/5
4. Login system question: 4/5
5. MAC unblock question: 1/5

Baseline Total: 15/25
Baseline Average: 3.00/5

Errors

1. **File not found (data.txt)**
    - **Issue**: Script failed when run from a different working directory.
    - **Solution**: Path handling updated using script-relative Path(__file__).resolve().parent.
2. **Broken/garbled answers**
    - **Issue**: Fixed character chunking split words/sentences, causing noisy retrieval context.
    - **Solution**: Replaced with word-aware chunking and used top-2 chunk retrieval.
3. **Invalid score input**
    - **Issue**: Values like 6 were accepted, breaking the 1-5 rubric.
    - **Solution**: Added strict input validation loop to accept only integers from 1 to 5.
4. **API key configuration issue**
    - **Issue**: Hardcoded/invalid API key caused generation failures.
    - **Solution**: Added environment-based key loading (GEMINI_API_KEY) and safe fallback.
5. **Unstable answer quality due to mixed context**
    - **Issue**: Retrieved context sometimes included partially unrelated data.
    - **Solution**: Tuned chunk size and benchmarked baseline vs tuned configuration as required.

Output

Questions Used

1. I'm seeing Error XV-505 in my terminal. What is the root cause and how do I fix it?
2. My system is extremely laggy. Which shards should I be worried about and what is the specific command to reset them?
3. Can I bypass the Theta-Sync requirement to install this on my old i7 laptop?
4. How does the OS handle user login now that passwords and 2FA are deprecated?
5. I have a MAC address that was black-holed by the Wraith Firewall. Can I unblock it using the Galactic-Relay?

```
PS D:\Daily_Tasks\Day10_Task> python index.py
r a RAM failure, but it is
Enter groundedness score (1-5): 1

Question: My system is extremely laggy. Which shards should I be worried about and what is the specific command to reset them?
Answer: ● vortex --realign-pulse: The primary fix for "System Lag" or "UI Ghosting." This command forces a hard reset of Shard-07 and Shard-04. ● vortex --deploy-shield: M
anually cycles the Nebula-Shield

"Quantum-Coherence." 1.2 Shard-Specific Responsibilities Each of the 12 shards is hard-coded to manage specific system silos to prevent resource contention: ● Shards 01-03
: Hardware Abstraction Layer
Enter groundedness score (1-5): 2

Question: Can I bypass the Theta-Sync requirement to install this on my old i7 laptop?
Answer: System Component Technical Description Required Resolution Protocol XV-909 CPU Architecture Theta-Sync Mismatch: The installer detected a non-Vortex processor (x86
/ARM64) lacking the required

processor that lacks the Theta-Sync instruction set, it will immediately terminate the installation and display Error Code: XV-909. Many junior technicians mistake this fo
r a RAM failure, but it is
Enter groundedness score (1-5): 3

Question: How does the OS handle user login now that passwords and 2FA are deprecated?
Answer: alphanumeric passwords, 2-Factor Authentication (2FA) via SMS, and standard optical fingerprint scanners. These methods were deemed "low-resonance" and vulnerable
to "Ghost-Admin" takeovers. 3.2

drive space for user data and applications. PAGE 3: THE AEGIS SECURITY LAYER & BIOMETRIC FREQUENCY 3.1 Deprecation of Legacy Authentication As of Version 9.2, NexusVortex
has removed all support for
Enter groundedness score (1-5): 4

Question: I have a MAC address that was black-holed by the Wraith Firewall. Can I unblock it using the Galactic-Relay?
 hardware failure. Contact support via Galactic-Relay Frequency 442.9 MHz. XV-505 Environment Path Configuration Error: The VORTEX_PATH variable is incorrectly mapped to t
he /vortex partition. Re-map VORTEX_PATH to /pulse/bin. Verify mapping with vortex --status. 5.4 Looking Ahead: Version 10.0 ("Singularity") The upcoming "Singularity" upd
ate (Q4 2026) will remove all kernel-level drivers for physical keyboards and mice. The OS will transition to a 100% "Neural-Link" and "Vocal-Gesture" API interface.

short-wave radio at 14.2 KHz is monitored by autonomous support agents. 5.3 Error Code Diagnostic Reference Table Error Code System Component Technical Description Require
d Resolution Protocol XV-909 CPU Architecture Theta-Sync Mismatch: The installer detected a non-Vortex processor (x86/ARM64) lacking the required hardware-level instructio
n set. Upgrade hardware to a Vortex-Gen2 or higher processor with native Theta-Sync support. XV-402 Storage Partition Pulse-Mirror Inconsistency: The /echo recovery partit
ion size deviates from the mandatory 512MB requirement. Re-partition the drive. Ensure
Enter groundedness score (1-5): 6
Invalid score. Please enter a number from 1 to 5.
Enter groundedness score (1-5):
```